

Manuel Bonilla López, PhD

Technical Program & Systems Leader

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[Personal Website](#) | [LinkedIn Profile](#) | [Google Scholar](#)

EXECUTIVE PROFILE

Technical program and systems leader with a Ph.D. in Physics and 16+ years of experience across industry and research spanning defense and commercial semiconductor manufacturing, microelectronics, digital manufacturing, Manufacturing Execution Systems (MES), optics, thin films, materials science, magnetism, and engineering data systems. Combines hands-on process and equipment engineering with multi-site implementation leadership.

Known for turning difficult operational problems into governed, sustainable systems: recovering stalled implementations, stabilizing manufacturing processes, building quality and data frameworks, coordinating cross-functional teams, and creating the work instructions, training, ownership models, schedules, and metrics required for long-term execution.

5	250–300	~50	~2,500
Sites Supported	Direct MES Users Supported	Tools Covered in Current Scope	Citations from 10 publications

LEADERSHIP VALUE

- **Program and systems leadership:** Connects business priorities, manufacturing needs, technical requirements, developers, information technology, quality, and operations across multiple sites.
- **Ambiguity-to-execution:** Turns unclear problems into roadmaps, accountable workstreams, requirements, schedules, decision structures, metrics, and sustainment plans.
- **Team enablement:** Assigns and tracks work, develops training syllabi and work instructions, prepares local administrators, and builds practical governance for adoption.
- **Technical depth:** Brings direct experience in semiconductor processing, optical coatings, equipment reliability, quality engineering, materials science, magnetism, statistical analysis, and automation.

CORE CAPABILITIES

Leadership & Operations	MES & Digital Manufacturing
Digital transformation Multi-site implementation Roadmaps Cross-functional leadership Stakeholder alignment Vendor coordination Capacity planning	MES architecture Digital travelers Routes/workflows WIP tracking Data collection SPC integration Document control Roles & permissions JIRA UAT
Manufacturing & Process Engineering	Quality & Process Control
Process ownership & development Production support Preventive maintenance Yield improvement DFM	RCCA PFMEA SPC OCAP Change control Defect taxonomy Quality workflows Corrective actions Risk assessment Process controls
Semiconductor, Deposition & Vacuum	Data, Automation & Advanced Metrology
ALD MBE E-beam Sputtering PECVD CVD Photolithography High/UHV vacuum Leak diagnosis Chamber recovery Tool ownership Thin-films	Python SQL R Power BI Data pipelines Dashboards Real-time monitoring Spectrophotometry Ellipsometry STM XPS/UPS AFM/SEM

PROFESSIONAL EXPERIENCE

L3Harris

Palm Bay, Florida

Senior Manufacturing Engineer

Mar 2024 – Present

– Multi-Site & Segment Manufacturing Systems Leadership

Nov 2025 – Present

- Lead and advise manufacturing-system implementation across Palm Bay, Tempe, Mason, Londonderry, and State College, aligning site roadmaps, configuration strategy, governance, process ownership, adoption, and sustainment.
- Support Eyelit MES adoption for semiconductor, power-supply, and related manufacturing in Palm Bay, Tempe, Mason, and Londonderry, including workflow design, administrator enablement, deployment planning, and best-practice sharing.
- Plan implementation activities for a separate Palm Bay Circuit Card Assembly organization of approximately 50–100 MES users, defining task ownership, sequencing, milestones, dependencies, and readiness actions.
- Serve as manufacturing-system lead for the State College CCA organization using Zeus, an internally developed solution; author work instructions, train the team, and coordinate requirements and issue resolution with developers and information technology.
- Build Microsoft Planner and supporting tracking structures that convert implementation roadmaps into assigned tasks, accountable owners, schedules, status visibility, risks, and follow-up actions.
- Develop training syllabi and deliver role-based instruction for functional users and prospective local administrators on routes, workflows, data collection, document control, SPC concepts, roles, troubleshooting, and governance.
- Contribute manufacturing context and site-deployment lessons to the segment-level Solumina/iBaseT digital-transformation program across multiple business areas.

– Department Digital Transformation & Manufacturing Execution

Mar 2025 – Nov 2025

- Assumed functional ownership of a stalled MES implementation and led recovery from paper- and Word-based travelers to governed digital manufacturing execution.
- Architected and deployed Eyelit as the foundational MES platform for the Microelectronics Department, supporting an initial population of approximately 200 users across operations, engineering, quality, metrology, and support functions.
- Translated manufacturing processes into digital routes, workflows, decision points, data-collection structures, user roles, process controls, work-in-process visibility, and document-control requirements.
- Implemented digital travelers integrating recipes, parameters, inspections, execution logic, and quality records to improve traceability, standardization, and production visibility.
- Designed functionality supporting production tracking, quality data collection, SPC, document control, tool-status visibility, preventive-maintenance alignment, and operational reporting.
- Authored functional documentation, work instructions, training materials, process flows, UAT scenarios, and issue-tracking structures; coordinated enhancements and troubleshooting through JIRA, vendor sessions, developers, and information technology.
- **Received a 2025 L3Harris R.I.S.E. recognition** for successfully implementing Eyelit MES, developing automated visualization platforms, and driving project adoption and sustainability.
- **Selected as a 2025 SAS Operations Rising Star Award winner**, recognizing exceptional talent, initiative, rapid learning-to-action, and tangible results within the first three years at L3Harris.

– Manufacturing Systems Foundations, Process Governance & Operations Support

Mar 2025 – Nov 2025

- Led department-wide Process Failure Mode and Effects Analysis (PFMEA) across approximately 50 tools spanning photolithography, deposition, wet and dry etch, chemical processing, chemical-mechanical planarization, laser engraving, lift-off, microscopy, and metrology.
- Documented hundreds of tool- and process-specific failure modes, effects, causes, prevention and detection controls, risk rankings, recommended actions, ownership, and completion evidence.

- Translated PFMEA findings into the department defect taxonomy, defect-code list, failure classifications, severity scale, occurrence and detection criteria, and standardized resolution logic, while identifying risks that could not yet be prevented or measured.
- Defined the department-wide Out-of-Control Action Plan (OCAP) response model connecting SPC triggers, tool shutdown and logging, wafer disposition, engineering ownership, nonconformance escalation, Root Cause and Corrective Action (RCCA), PFMEA updates, and closure requirements.
- Designed and institutionalized the department's first end-to-end SPC and analytics framework, authoring control plans, standards, review cadence, metrics governance, dashboards, automated notifications, and user training.
- Implemented environmental health monitoring using relative-humidity and temperature loggers, ISO-based alarm limits, automated Python pipelines, live dashboards, and Microsoft Teams notifications.
- Led development of the equipment and facility preventive-maintenance calendar by reviewing manuals, interviewing tool owners and equipment personnel, and converting required activities and frequencies into an annual schedule.
- Built the facilities workload model that supported approval of dedicated building support; served on the Change Control Board and partnered with deposition engineering on e-beam chamber improvements, revised crystal monitoring, and operational controls.
- Supported equipment readiness by leading safety risk assessments and developing the department preventive-maintenance framework with equipment personnel, including defining required PM activities and converting them into an annual scheduling and visibility system; *received two 2024 L3Harris R.I.S.E. recognitions* for safety-assessment leadership and preventive-maintenance / operational visibility contributions.

Lockheed Martin

Orlando, Florida

Senior Optical Manufacturing Process Engineer

Jan 2022 – Jan 2024

- Served as coating-area subject-matter expert for Missiles and Fire Control, providing production support to approximately eight process engineers and 30 operators across coating production, process engineering, equipment readiness, and coating design.
- Led the oxide-coating engineering team as technical contact, coordinating approximately four engineers and overseeing visible-spectrum chambers, processes, operators, and priorities while providing cross-support to the separate infrared-coating team.
- Supported F-35, Sniper Advanced Targeting Pod, Javelin, Joint Air-to-Ground Missile, Modernized Day Sensor Assembly, IRAD initiatives, and additional non-recurring programs.
- Led Independent Research and Development for diamond-like carbon (DLC) and oxide-based coating systems and directed antireflective, high-reflective, and environmentally durable thin-film production across visible and infrared applications.
- Owned the lifecycle of customer-requested, IRAD, and non-recurring engineering work: optical modeling, budgeting, material selection and purchasing, equipment setup, recipe development, acceptance criteria, production execution, and shipment support.
- Designed coating processes to customer drawings, program specifications, MIL-PRF-13830 requirements where applicable, and internal production controls.
- Owned Tecport Optics and Bühler Leybold PVD/CVD systems using e-beam evaporation, sputtering, thermal evaporation, and plasma-enhanced chemical vapor deposition.
- Diagnosed coating and equipment issues affecting yield, durability, and reliability, including delamination, non-uniformity, spectral deviation, environmental-test failures, chamber leaks, witness-sample absorption, and temperature-calibration errors.
- Applied structured RCCA to reduce defective coating runs from multiple failures per week to near zero within months and enable delivery up to one month early on select programs.
- Recovered an environmental humidity test chamber and a plasma-enhanced chemical vapor deposition system used for DLC coatings; led corrective actions on supplier and vendor quality gaps.
- *Received four NextGen Recognition Awards* for collaboration, yield improvement, and root-cause resolution, *plus the Q4 2022 PMT Advocate Award* for continuous improvement and positive manufacturing culture.

Intel Corporation

Hillsboro, Oregon

Semiconductor Process Engineer

Jun 2020 – Jul 2021

- Supported high-volume front-end semiconductor manufacturing at Intel's D1 fabrication facility as atomic layer deposition (ALD) tool owner, driving equipment reliability, film quality, process stability, troubleshooting, and technology-transition support.
- Owned two ALD manufacturing tools comprising four deposition sub-chambers for high-k dielectric film deposition supporting FinFET transistor production.
- Improved tool availability and process stability using design of experiments, SPC, chamber-pressure diagnostics, preventive-maintenance optimization, and pump-life analysis.
- Coordinated directly with equipment vendors to sustain high-volume production and resolve equipment-performance issues.
- Led conversion of two tools from a previous product line to the current production flow, supporting continuity during technology transitions.
- Restored inactive tools to production through coordinated diagnostics, vendor support, chamber recovery, and production qualification.

RESEARCH AND TECHNICAL EXPERIENCE

University of South Florida

Tampa, Florida

Research Associate — Advanced Materials Physics

Aug 2013 – Jun 2020

- Led the experimental work enabling the first reported room-temperature ferromagnetism in monolayer VSe₂, establishing a widely cited result in two-dimensional materials and magnetism.
- **First author:** ["Strong Room-Temperature Ferromagnetism in VSe₂ Monolayers on van der Waals Substrates,"](#) Nature Nanotechnology 13, 289–293 (2018).
- **Independent editorial highlight:** ["2D Magnetism Gets Hot,"](#) Nature Nanotechnology (2018).
- Synthesized and characterized monolayer VSe₂ and developed deposition recipes for TiSe₂, MoSe₂, WSe₂, and TaSe₂ to study magnetic states, charge-density waves, excitonic behavior, band structure, doping, and surface morphology.
- Specialized in molecular beam epitaxy, physical vapor deposition, ultra-high-vacuum systems, scanning tunneling microscopy and spectroscopy, angle-resolved photoemission spectroscopy, vibrating-sample magnetometry, Raman spectroscopy, atomic-force microscopy, X-ray techniques, and electrical characterization.
- Installed, configured, maintained, and commissioned ultra-high-vacuum systems, including a shared departmental molecular beam epitaxy platform supporting multi-user research operations.
- Designed end-to-end experimental workflows and data-analysis systems and built predictive models linking process conditions to electronic, magnetic, structural, and surface behavior.
- Selected as guest scientist at **SLAC National Accelerator Laboratory in California** and **SOLEIL Synchrotron in France** for advanced characterization studies.

University of Puerto Rico

Humacao, Puerto Rico

Research Associate — Nanomaterials, Polymer Electronics & Sensor Devices

Aug 2010 – Aug 2013

- Conducted research in polymer electronics, nanofiber sensors, microfabrication, electrical characterization, and defense-related smart-textile concepts through PREM collaborations with the University of Pennsylvania and University of Nebraska–Lincoln.
- Fabricated gas-sensing diodes and transistors by electrospinning polymer nanofibers directly onto silicon wafers to form active sensing layers for chemical detection.
- Deposited metal electrodes by thermal evaporation and executed current–voltage and gas-response testing to evaluate sensitivity, stability, and performance.
- Performed structural and morphological analysis using scanning electron microscopy to validate nanofiber formation and device architecture.

- Developed polymer formulations and nanofiber recipes to tune chemical composition and improve gas-sensing response.
- Supported smart-textile concepts integrating sensing devices into fabrics for environmental and defense-related gas detection.
- Contributed to carbon-nanotube sensing platforms, including early non-invasive glucose-detection development using chemical vapor deposition and photolithography.

AWARDS & RECOGNITION

- **L3Harris Rising Star Award (2025):** Selected as a SAS Operations Rising Star, recognizing exceptional talent, initiative, rapid contribution, and tangible results within the first three years at L3Harris.
- **Lockheed Martin PMT Advocate Award (4th Quarter 2022):** Recognized for championing continuous improvement and improving operator-management communication and daily issue resolution.
- **Lockheed Martin Next Gen Recognition Awards:** Four awards recognizing cross-functional collaboration, yield improvement, root-cause resolution, and operational impact.
- **L3Harris R.I.S.E. Awards (3):** Recognized for leading accelerated equipment Risk Safety Assessments; developing a preventive-maintenance scheduling and real-time visibility solution with the equipment team; and successfully implementing Eyelit MES with automated visualization platforms, adoption, and sustainment.

EDUCATION & ACADEMIC RECOGNITION

- **Ph.D., Physics** | University of South Florida
- **M.Sc., Physics** | University of South Florida
- **B.Sc., Physics Applied to Electrical Engineering** | University of Puerto Rico
- **Coursework concentration in Computational Mathematics** | University of Puerto Rico
- Florida Louis Stokes / LSAMP graduate fellowship
- American Physical Society Bridge Program Physics Graduate Fellow — inaugural cohort, University of South Florida.

CERTIFICATIONS & CONTINUING EDUCATION

- **Google Data Analytics Professional Certificate** | Google via Coursera
- **Leading People and Teams Certificate — In Progress** | University of Michigan via Coursera
- **CS50 Computer Science Coursework — In Progress** | Harvard University via edX

TEACHING & TRAINING

- **Graduate Teaching Associate** | Physics Department, University of South Florida — developed and delivered Physics I and II laboratory instruction, experimental setup, technical explanations, grading, and laboratory safety.
- **Academic Tutor** | University of Puerto Rico — supported undergraduate Calculus I–II, Precalculus I–II, and Physics I–II through university tutoring services.
- **Manufacturing and systems trainer** | Develops administrator curricula, role-based training, work instructions, process flows, and adoption materials for manufacturing users, engineers, and support teams.

RESEARCH IMPACT & EXTERNAL RECOGNITION

- **Publication record:** 10 peer-reviewed publications with approximately 2,500 scientific citations | [Google Scholar](#)
- **Career profiles:** [APS News Bridge Program graduate profile](#) (2020) | [Symmetry Magazine PhD pathway feature](#) (2015).
- **Research and community features:** [USF feature on atomically thin magnetic semiconductors](#) (2020) | [El Oriental — “¡Orgullo Boricua!” doctoral profile](#) (July 29, 2020, p. 12).